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Sentiment Analysis

Sentiment Analysis using NLTK:

Sentiment analysis is a technique which detects the underlying sentiment on specific textual content. It is considered as a process of classifying text on the basis of labels i.e. positive, negative or neutral.

To perform Sentiment Analysis using NLTK, it has a supportable package named as VADER**. VADER stands for Valence Aware Dictionary and Sentiment Reasoner.**

```
import nltk
nltk.download('vader_lexicon')
```

Example 1:

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
vader_analyzer=SentimentIntensityAnalyzer()
text1="I am feeling good" # The text is positive.
print(vader_analyzer.polarity_scores(text1))
```

Output :
{'neg': 0.0, 'neu': 0.185, 'pos': 0.815, 'compound': 0.5267}

Here the output is giving some labels:

- **neg : Negative**
- **neu : Neutral**
- **pos : Positive**

It has given 'pos' value as 0.815 which is maximum of all the other values since the statement is positive. Similarly, we can check it on other emotions as well.

Example 2:

```
text1="I hate tea"
print(vader_analyzer.polarity_scores(text1))
```

Example 3:

```
text="I like Data Analytics"
result=vader_analyzer.polarity_scores(text)
# To find percentage of ratings
print("The sentence is rated as ",result['pos']*100,"% Positive")
print("The sentence is rated as ",result['neg']*100,"% Negative")
print("The sentence is rated as ",result['neu']*100,"% Neutral")
```

```
if result['compound']>=0.05:
    print("Overall rating for sentence is Positive")
elif result['compound']<=-0.05:
    print("Overall rating for sentence is Negative")
else:
    print("Overall rating for sentence is neutral")
```

Example 4: (Set –A)

Consider the following review messages. Perform sentiment analysis on the messages.

i. I purchased headphones online. I am very happy with the product.
ii. I saw the movie yesterday. The animation was really good but the script was ok.
iii. I enjoy listening to music
iv. I take a walk in the park everyday